MIL-C-14039C
18 November 1981
SUPERSEDING
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19 April 1976

MILITARY SPECIFICATION

COMPASS, SUN, UNIVERSAL TYPE:

0 TO 90 DEGREES, NORTH AND SOUTH LATITUDES,

WITH CARRYING CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sun compass for use in any latitude north or south of the equator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6605

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-3918	- Lubricating Oil, Instrument, Jewel Bearing.
MIL-G-23827	- Grease, Aircraft and Instrument.
MIL-C-83488	- Coating, Aluminum, Ion Vapor Deposited.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13216E8350	- Compass, Sun, Universal, 0° to 90°, North and South Latitudes, with Case.

(Copies of specifications, standards, and drawings required by contractors in connection with specific aquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The sun compass (hereinafter called "compass") with case, and tools and accessories with case shall be in accordance with Top Assembly TA13216E8350 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the compass, compass case, accessories, tools and accessory case. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (first produced compass). The contractor shall furnish one or more compasses, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce compasses that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the government (see 6.3).

3.3 Material. Material shall be as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Environmental conditions.

3.4.1 Operating temperature. The compass shall perform as specified herein in an ambient temperature from minus 25° F to plus 110° F.

3.4.2 Storage temperature. The compass shall not be damaged by storage in an ambient temperature of minus 28° F, plus or minus 3° F, for a period of 6 hours, nor in an ambient temperature of plus 145° F. plus 6° F, minus 0° F, for 4 hours, with induced relative humidity of from 10 to 85 percent.

3.5 Operational performance.

3.5.1 Mechanical operating conditions. All mechanical features except the clock shall operate at an ambient temperature from minus 25° F, plus 0° F, minus 3° F, for 6 hours to plus 110° F, plus 0° F, minus 5° F, for 1 hour. There shall be no binding of moving parts in any feature of the unit.

3.5.2 Clock assembly. The clock mechanism shall be designed to withstand severe use and extreme temperatures. There shall be no evidence of backlash in the clock case and bearing assembly or the horizontal carrier assembly.

3.5.2.1 Warm operational. The clock mounted in the compass shall neither gain nor lose more than 2 minutes in any 24-hour period when operated at plus 70° F, plus or without being rewound.

3.5.2.2 Hot operational. The clock shall remain operational when subjected to solar radiation of 355 British thermal units/square foot/hour, (Btu/sq ft/hr) at a temperature of 110° F, plus 0° F, minus 5° F, for not less than 1 hour. The clock shall neither gain nor lose more than 24 seconds during this period.

3.5.2.3 Cold operational. The clock, mounted in the compass, shall operate without interruption in an ambient temperature of not more than minus 25° F, plus 0° F, minus 3° F, without benefit of solar radiation for 6 continuous hours. It shall neither gain nor lose more than 30 seconds during the 6-hour period.

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3.5.2.4 Hot storage. The clock shall remain operational when subjected to a temperature of not less than plus 145° F, plus 6° F, minus 0° F, for a period not less than 4 hours. The clock shall neither gain or lose more than 20 seconds during this period.

3.5.2.5 Cold storage. At temperatures lower than minus 28° F, the clock need not continue to run, but the hands must remain sufficiently free to permit manual setting.

3.5.3 Accuracy of gnomon rod and star sight. The combined root mean square error of the declination scale, the eccentricity of the gnomon rod and star sight shall not exceed plus or minus 10 minutes of arc.

3.5.4 Latitude and micrometer scales. The latitude and micrometer scales shall have a root mean square error not more than plus or minus 10 minutes. When the latitude scale is set at 90 degrees North and the micrometer scale is set at zero, the clock shall not be out of the horizontal plane by more than plus or minus 20 minutes.

3.5.5 Level vial. The level vial shall have a sensitivity not more than 20 minutes of arc per 2 millimeters (mm) displacement. The liquid in the vial shall have a freezing point lower than minus 25° F and shall neither boil nor deteriorate up to a temperature of plus 110° F, plus 0° F, minus 4° F. A spare vial assembly shall be furnished with each compass.

3.5.6 Azimuth accuracy. The root mean square error of the azimuth readings of the compass, compared with an established azimuth accurate to 2 minutes of arc, shall not exceed plus or minus 1 degree. No single error shall exceed plus or minus 2 degrees.

3.5.7 Instrument case waterproofness. The instrument case shall be waterproof to the extent that not more than 20 cubic centimeters (cc) of water shall enter the case when tested as specified in 4.5.2.12.

3.6 Charts and tables. All charts and tables shall be as shown on the drawings. All printing shall be accurate and legible.

3.6.1 Sun time correction chart. The data columns on the card and the data columns on the slide shall be in precise alinement on both sides of the card and throughout the range of travel of the slide.

3.6.2 Hour-angle star chart. The pivot shall be accurately centered in the disks, mounted securely to prevent working loose in normal use, and shall permit the disks to turn freely. The index lines for the stars shall be plotted at the angular intervals shown in table I proceeding clockwise from the December 31 graduation. Graduations and index lines shall be within plus or minus 5 minutes of arc of the true position they represent.

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TABLE I. Plotting data for hour-angle star chart.

Star	Angle measured from December 31		Star	Angle measured from December 31	
	Degrees	Minutes		Degrees	Minutes
Canopus	3	52	Vega	180	47
Capella and Rigel	21	26	Shaula	197	02
Aldebaran	31	19	Antares ...	213	00
Hamal	68	30	Arcturus ..	246	14
Gamma (γ) Cassiopeia.	86	10	Spica	258	57
Deneb Kaites	89	19	Acrux	273	38
Alpheratz	98	12	Denebola ...	282	57
Markab	114	01	Dubhe	294	25
Fomalhaut	115	52	Regulus	308	09
Deneb	149	40	Pollux	344	01
Altair	162	30	Procyon	345	24
			Sirius	358	50

3.7 Operating tools and accessories. Tools and accessories shall be provided and placed in the accessory case as shown on the assembly drawings.

3.8 Lubrication. All moving parts of the compass, with the exception of the clock assembly, shall be lubricated with grease conforming to MIL-G-23827. The clock assembly shall be lubricated with oil conforming to MIL-L-3918.

3.9 Identification marking. Each compass and case shall be identified in accordance with MIL-STD-130.

3.10 Treatment and painting. Treatment and painting shall be as specified on the applicable drawings. When not specified, treatment and painting shall be in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.11 Protective coating. Protective coating for all steel parts shall be as specified on the applicable drawings. Where coating is not specified, all steel parts with wearing surfaces, except bolts, nuts, taper pins, dowel pins, and those parts designated as CRES (corrosion resistant steel), shall be chromium plated. All other steel parts shall be cadmium plated conforming to QQ-P-416, class 2, type II or coated in accordance with MIL-C-83488, class 2, type II.

3.12 Workmanship. The compass and accessories shall be finished without cracks or binds. Contact and bearing surfaces shall be true to allow for ease of fit and movement. All parts shall be free of dirt or abrasive material. Graduations and lettering shall be uniformly finished. The accessory case shall be stitched without any ruffing or crimping.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced compass shall be cause for disassembly, in the presence of a Government representative, of the first produced compass to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced compass. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First produced compass inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First produced compass inspection.

4.3.1 Examination. The first produced compass shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced compass shall be subjected to the tests marked "X" in column 2 of table II, with the tests conducted in the order listed. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table II, with the tests conducted in the order listed. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The compass shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Dimensions not as specified.
103. Components missing or not as specified.
104. Level vial cracked or improperly seated.
105. Sun time correction chart not as specified.
106. Hour-angle star chart not as specified.
107. Operating tools and accessories missing or not as specified.
108. Lubrication not as specified.
109. Steel parts not protective coated as specified.
110. Workmanship not as specified.

Minor

201. Identification marking, missing, incomplete, or illegible.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. The time standard of comparison shall be high quality comparable to that of a Navy chronometer and shall be clearly readable in seconds. The standard shall be compared at 24-hour intervals with Naval Observatory time during the testing periods. Any gain or loss of the standard shall be prorated linearly over the 24-hour period for purposes of determining corrections to Naval Observatory time. Readings of the standard and of the item undergoing test shall be taken at hourly intervals during all tests. The gain or loss of the test item with respect to the standard corrected to Naval Observatory time shall be computed at hourly intervals.

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4.5.2.2 Test schedule.

TABLE II. Test schedule.

Test sequence	Type of test: First produced compass	Type of test: Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6
-	X	X	Test conditions	4.5.2.1	--------
1	X	-	Warm operational	4.5.2.3	3.5.2.1
2	X	-	Hot operational	4.5.2.4	3.5.2.2
3	X	-	Cold operational	4.5.2.5	3.5.2.3
4	X	-	Hot storage	4.5.2.6	3.5.2.4
5	X	-	Cold storage	4.5.2.7	4.5.2.5
6	X	X	Gnomon rod and star sight	4.5.2.8	3.5.3
7	X	X	Latitude and micrometer scales	4.5.2.9	3.5.4
8	X	-	Level vial	4.5.2.10	3.5.5
9	X	X	Azimuth accuracy	4.5.2.11	3.5.6
10	X	-	Case submersion	4.5.2.12	3.5.7

4.5.2.3 Warm operational. At plus 70° F, plus or minus 5° F, level the compass and set the latitude to 90 degrees North. Completely wind and set the clock with either gnomon rod attached. Bending of moving parts, evidence of backlash, gain or loss of more than 2 minutes in any 24-hour period, or failure of the clock to run continuously for not less than 120 hours without rewinding shall constitute failure of this test.

4.5.2.4 Hot operational. Wind and accurately set the clock. Level the compass with gnomon rod attached and set the latitude to zero degrees. the compass shall be temperature-stabilized at plus 110° F, plus 0° F, or minus 5° F, and a source of directed radiant energy simulating solar radiation of 355 Btu/sq ft/hr shall be applied to the instrument for not less than 1 hour. A suitable source for this energy is five 300-watt R-40 reflector spot incandescent lamps (or equal) mounted on 6-inch centers in the form of a cross with one lamp on the center common to both limbs. Such a source, with the plane of the cross horizontal, mounted directly over the compass at a hight of 38 inches from the tip of the center lamp to the top of the sighting bar assembly will give the required radiant flux to a close approximation. Evidence of binding of moving parts or gain or loss of more than 24 seconds during the 4-hour test period shall constitute failure of this test.

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4.5.2.5 Cold operational. Wind and accurately set the clock. Level the compass with gnomon rod attached and set the latitude to 80 degrees North. The compass shall be temperature-stabilized at not more than minus 25° F without benefit of solar radiation for not less than 6 hours. Evidence of binding of moving parts or gain or loss of more than 30 seconds during the 6-hour period shall constitute failure of this test.

4.5.2.6 Hot storage. Wind and accurately set the clock. Level the compass with gnomon rod attached and set the latitude to zero degrees. The compass shall be temperature-stabilized at not less than 145° F, plus 6° F, minus 0° F, for 4 hours. Evidence of binding of moving parts or gain or loss of more than 20 seconds during the 4-hour period shall constitute failure of this test.

4.5.2.7 Cold storage. Level the compass with a gnomon rod attached and set the latitude to 80 degrees North. Subject the compass to a temperature of not more than minus 28° F for a continuous period of 6 hours. At the conclusion of the exposure period, examine the clock. Binding of moving parts or inability of the hands of the clock to be set manually to desired time settings shall constitute failure of this test.

4.5.2.8 Gnomon rod and star sight accuracy. The gnomon rod and star sight shall be tested by comparison with an established 20 degree angle between two collimators. Read 20 degree increments of the declination scales estimating to one-tenth of a division. An example of the test is tabulated in table III. Computed root mean square error exceeding plus or minus 10 minutes of arc shall constitute failure of this test.

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TABLE III. Gnomon rod and star sight, sample test.

Collimation values in degrees	Gnomon Rod values in degrees	Difference in degrees [(b)-(a)]x2	Difference in minutes	Squared differences VV = (d)2
(a)	(b)	(c)	(d)	(e)
90.0	90.0	-	-	-
70.0	70.1	0.2	12	144
70.0	70.0	-	-	-
50.0	49.9	-0.2	-12	144
50.0	50.0	-	-	-
30.0	30.0	0.0	0	0
30.0	30.0	-	-	-
10.0	10.1	0.2	12	144
10.0	10.0	-	-	-
-10.0	-10.0	0.0	0	0
-10.0	-10.0	-	-	-
-30.0	-30.1	0.2	12	144
-30.0	-30.0	-	-	-
-50.0	-50.0	0.0	0	0
-50.0	-50.0	-	-	-
-70.0	-69.9	-0.2	-12	144
-70.0	-70.0	-	-	-
-90.0	-90.0	0.0	0	0
				Σ =VV 720

Root mean square error = $\varepsilon = \pm \sqrt{\frac{\Sigma VV}{n-1}}$

$= \pm \sqrt{\frac{720}{8}}$

$= \pm$ 9.5 minutes

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4.5.2.9 Latitude and micrometer scales. Level the compass and set the latitude scale to 90 degrees North with the micrometer scale set to zero. Place a level with a sensitivity of at least 5 minutes per 2 mm on the face of the clock normal to the horizontal axle of the clock. Determine if the latitude, index mark, 90 degree graduation mark, and micrometer scale are in proper relation to each other. Set the latitude scale to precisely 90 degrees North according to the adjacent index mark, and record the micrometer scale. An example of the test is tabulated in table IV. The clock out of the horizontal plane by more than plus or minus 20 minutes, or the computed root mean square error of more than plus or minus 10 minutes shall constitute failure of this test.

TABLE IV. Latitude and micrometer scales, sample test.

Lat. scale degrees	Micrometer scale degrees	Lat. scale degrees	Micrometer scale degrees	Difference (minutes)		VV	
				N	S	N	S
				V	V		
90.0N	0.2	10.0S	0.0	12	0	144	0
80.0	0.0	20.0	0.0	0	0	0	0
70.0	9.8	30.0	0.3	-12	+18	144	324
60.0	0.1	40.0	9.8	6	-12	36	144
50.0	9.9	50.0	0.2	- 6	12	36	144
40.0	0.0	60.0	0.1	0	+ 6	0	36
30.0	0.2	70.0	9.8	12	-12	144	144
20.0	9.8	80.0	9.9	-12	6	144	36
10.0N	0.1	90.0S	0.0	+ 6	0	36	0
0.0	9.9	-	-	- 6	-	36	-
						Σ = 1548	

Root mean square error = $\epsilon = \pm\sqrt{\frac{\Sigma VV}{n-1}} = \pm\sqrt{\frac{1548}{18}} = \pm 9.3$ minutes

4.5.2.10 Level vial. The level vial shall be subjected to the temperatures specified in 3.5.5. Any evidence of freezing, boiling, or deterioration of the liquid shall constitute failure of the test. The sensitivity of the level shall be tested on a level trier or similar hinged-beam micrometer instrument. Sensitivity of more than 20 minutes of arc per 2-mm displacement shall constitute failure of this test.

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4.5.2.11 Azimuth accuracy. The compass shall be tested by field observing on the sun with the gnomon rods and observing on the stars with the star sight. At least one azimuth shall be determined in each quarter of the 24-hour day, with not less than 2-hour intervals between observations. No less than four determinations shall be made with the gnomon rod, and no less than two shall be made with the star sight. Determined azimuths differing by more than plus or minus 1 degree root mean square error, or a single error more than plus or minus 2 degrees from the established azimuth, shall constitute failure of this test. A suggested procedure of test and tabulations follows:

Sample test

To perform this test accurately, lay off four reference marks from the initial azimuth in a clockwise direction in increments of 45 degrees. Place the reference marks at a radius of at least 100 feet from the initial point of the rays. Wind, adjust, and level the compass over the initial point with the gnomon rods attached. Sighting through the open sights, carefully aline the stake representing the established azimuth. Holding this line-of-sight by clamping the upper motion screw, turn the inner circle to aline the sun with the gnomon rod. When the sun is accurately positioned on the screen, read the azimuth of the compass to tenths of minutes and record the column 2 adjacent to zero degrees (the assumed value of the initial azimuth) as illustrated in the sample test form, table V. Unclamp the upper motion screw, sight in stake No. 2 at the 45 degree interval, and reclamp. Rotate the clock assembly and again center the sun upon the screen. Record the compass reading in column 2a adjacent to the 45 degree value in column 1. Repeat this procedure of sighting and observing for each 45 degree increment until reaching the 180 degree marker. At this point look through the sights backwards, unclamp the upper motion screw and sight-in stake No. 2. Aline the sun on the screen as before. Continue reading and recording as cited above. A correction factor of 0.2 degree (caused by the eccentric sights) (column 2b) is considered in the sample test form. The correction factor is positive when the stakes are viewed through the sights forward and is negative when viewed through the sights backward. Column 3 is the difference between the established and the corrected observed azimuth. Column 4 is the squared values of column 3 from which the root mean square values are determined as shown in the formula.

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TABLE V. Sample test.

(1)	(2)		(3)	(4)
Established azimuth plus increments of 45 degrees	Observed azimuth (a) Uncorrected	(b) Corrected	V Column (1)-(2) (b)	VV
Degrees	Degrees	Degrees	Degrees	Degrees
0	1.4	1.6	-1.6	2.56
45	44.2	44.4	0.6	0.36
90	90.8	91.0	-1.0	1.00
135	134.4	134.6	0.4	0.16
180	180.3	180.1	-0.1	0.01
225	226.0	225.8	-0.8	0.64
270	270.7	270.5	-0.5	0.25
315	314.3	314.1	0.9	0.81
				Σ VV 5.79

Root mean square error = $\varepsilon = \pm \sqrt{\frac{\Sigma VV}{n-1}} = \pm \sqrt{\frac{5.79}{8-1}}$ $\pm$ 0.9 degrees

4.5.2.12 Case submersion. Submerge the empty instrument case in a container of water to a depth of not less than 30 inches above the top of the case for a period of not less than 2 hours. More than 20 cc of water entering the instrument case at the end of the 2-hour period shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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111. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
112. Fiberboard boxes not closed as specified for level A.
113. Consolidation not as specified.
114. Commercial packaging not as specified.
115. Strapping not as specified for level A.
116. Marking illegible, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Compass. The compass and accessories contained in the instrument case shall be preserved in accordance with method IIb of MIL-P-116, using the case as the inner container and a fiberboard box conforming to PPP-B-636, W5c as the outer container. The fiberboard box shall be closed in accordance with method V of the appendix to the box specification.

5.1.1.2 Technical publications. The technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.3 Accessories. The accessories contained in the accessory case and the preserved technical publications shall be placed in a fiberboard box conforming to PPP-B-636, W5c and closed in accordance with method V of the appendix to the box specification.

5.1.1.4 Consolidation. The preserved compass and accessories, including the technical publications, shall be placed in a box conforming to PPP-B-636, W5c and closed in accordance with method V of the appendix to the box specification.

5.1.2 Level B. The compass, technical publications, and accessories shall be preserved as specified for level A in 5.1.1, except that the compass and accessories contained in the instrument case shall be preserved in accordance with MIL-P-116, method IA-14.

5.1.3 Commercial. The compass and accessories shall be placed in their respective cases. The cases and technical publications shall be preserved in accordance with MIL-STd-1188.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

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5.2.1 Level A. The compass and accessories shall be packed in boxes conforming to PPP-B-601, overseas type, grade B, or PPP-B-621, class 2. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The compass and accessories shall be packed as specified in 5.2.1 for level A, except that the boxes shall be domestic type, and strapping shall be either finish A or B.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A and B. Level A and level B marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The compass is intended to be used as a vehicular-mounted, ground navigation instrument for establishing or determining true azimuths in all north and south latitudes, especially in such areas as desert or arctic wastes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Time frame required for submission of the first produced compass, and number of compasses required (see 3.2).

c. Color of paint required for compass and clock (see 3.10).

d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

e. When strapping shall be other than as specified (see 5.2.1).

6.3 First produced compass. Any changes or deviations of production compasses from the approved first produced compass during production will be subject to the approval of the contracting officer. Approval of the first produced compass will not relieve the contractor of his obligation to furnish compasses conforming to this specification.

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6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manuals to be furnished with each compass.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary, and instructions regarding shipment of compasses.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - MC

Preparing activity:
Army - ME

Project 6605-0312

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-14039C Compass, Sun, Universal Type: 0 to 90 Degrees, North and South Latitudes, with Carrying Case

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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